

KELASH KUMAR

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Summary

Computer Science graduate from Sukkur IBA University, focused on Data Engineering, with backend engineering experience and end-to-end portfolio pipelines spanning ETL/ELT, real-time streaming, Change Data Capture, and Medallion lakehouse architecture. Hands-on with Python, SQL, PostgreSQL, Airflow, dbt, Kafka/Redpanda, Debezium, FastAPI, and Docker. Every project is containerized, orchestrated, and publicly available on GitHub with full architecture docs.

Technical Skills

Languages & Query: Python, SQL, Bash/Shell, Java

Data Engineering: Apache Airflow, dbt, Kafka/Redpanda, Debezium, Great Expectations, Metabase, Pandas, ETL/ELT, Data Modeling, Data Warehousing, Medallion Architecture, CDC, SCD Type 2

Databases & Storage: PostgreSQL, MySQL, SQLite, DuckDB, MinIO (S3-compatible), Parquet

DevOps & Cloud: Docker, Docker Compose, Linux, Git, GitHub Actions, CI/CD, FastAPI, Supabase Auth (JWT), REST APIs, pytest, pre-commit, Makefile

Experience

Backend AI Engineering Intern

FlyRank AI

Jun 2026 – Aug 2026

- Built REST APIs in FastAPI with full CRUD, validation, and proper status codes; migrated persistence from in-memory to SQLite to Postgres (Docker) with zero changes to route/service logic.
- Added Supabase JWT authentication and reusable middleware for protected routes; containerized full stack with Docker Compose for one-command startup with persistent data.

Projects

Real-Time Crypto Streaming Pipeline

GitHub

Python, Redpanda, PostgreSQL, dbt, Airflow, Metabase, Docker, GitHub Actions

- Built a 9-service Docker Compose pipeline that ingests live prices for 10 crypto assets into Redpanda every ~60s (~14,400 updates/day); Python consumer handles malformed JSON and retries.
- Transformed data through a 3-layer dbt model (staging → intermediate → mart) on a 5-min Airflow schedule with a freshness gate, achieving sub-5-minute API-to-Metabase latency.
- Built a 4-job GitHub Actions CI/CD pipeline (~90s); 8 unit tests passing.

Change Data Capture Pipeline

GitHub

PostgreSQL, Debezium, Redpanda, Python, dbt, Airflow, Docker

- Cut replication lag from hourly-batch (~60 min) to <10s with log-based CDC via Debezium capturing Postgres WAL events into Redpanda, then an idempotent Python consumer writing to the warehouse via primary-key upserts, eliminating duplicates under at-least-once delivery.
- Built SCD Type 2 historical dimensions in dbt for point-in-time analytics on mutable source tables; isolated malformed events via a dead-letter queue so bad records never block the pipeline.

Medallion Data Lakehouse with Quality Gates

GitHub

MinIO, DuckDB, Pandas, Great Expectations, Airflow, Parquet

- Built a Bronze → Silver → Gold lakehouse on MinIO with Parquet and DuckDB as the SQL engine, enforcing 31 Great Expectations checks across layers to block bad partitions from promotion, with partition-aware Airflow DAGs for safe backfills.

Education

Sukkur IBA University

Sukkur, Pakistan

Bachelor of Science in Computer Science

Aug 2022 – May 2026

Relevant Coursework: Database Systems, Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering, System Design

Certifications

- IBM Data Engineering Professional Certificate — Coursera (2026)
- Snowflake Data Engineering Professional Certificate — Coursera (2026)
- Google Data Analytics Professional Certificate — Coursera (2025)

Achievements & Honors

National Skill Competency Test (NSCT), 2026 — Ranked in the 92.2 percentile nationally; top marks in Programming, Database, and AI/ML & Data Analytics.

Sindh Talent Hunt Program (STHP) Scholarship, 2022 — Fully-funded merit scholarship covering the BS Computer Science program at Sukkur IBA University.